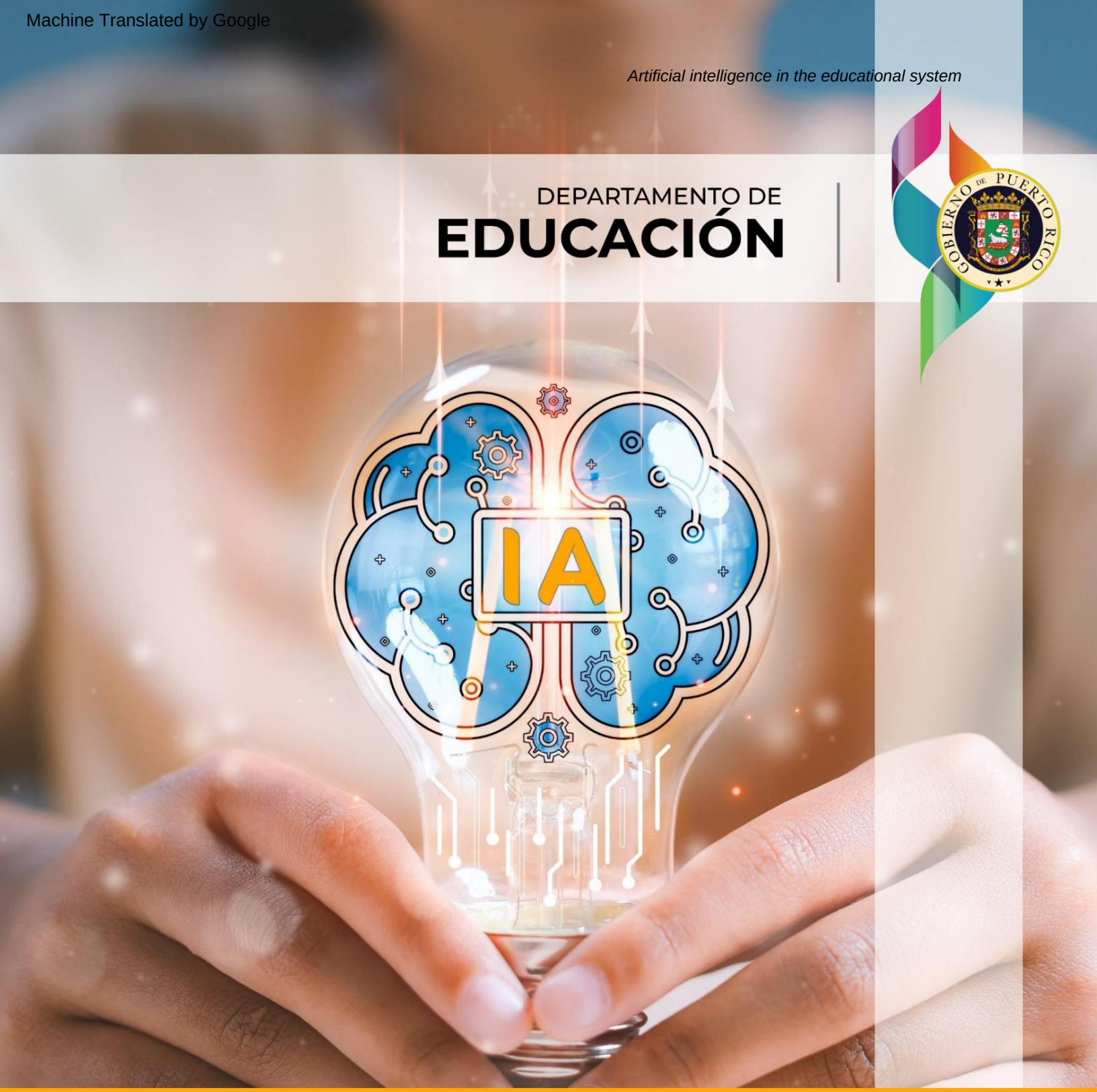
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DEPARTAMENTO DE
EDUCACIÓN



Artificial intelligence in the educational system

Undersecretary for Academic and Programmatic Affairs

2024-25

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Posted by

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DEPARTAMENTO DE
EDUCACIÓN

Artificial intelligence in the educational system

2024

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For legal purposes, in relation to the Civil Rights Act of 1984, the use of the terms school principal, teacher, students and any other terms that may refer to both genders, includes both male and female.

VALIDITY

This normative document will be in effect until the next curriculum review is carried out in accordance with the current School Curriculum Regulations of the Department of Education of Puerto Rico or its revision. This document repeals previous provisions or other established norms that contravene the content through public policy (circular letters, manuals, guides or memoranda) that are in conflict, in whole or in part.

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MESSAGE FROM THE SECRETARY

GOBIERNO DE PUERTO RICO
DEPARTAMENTO DE EDUCACIÓN
Secretaria | Dra. Yanira I. Raíces Vega | raices_y@de.pr.gov

NUEVAS HERRAMIENTAS CURRICULARES PARA LOS ESTUDIANTES, LOS MAESTROS Y LAS FAMILIAS COMO PARTE DEL LEMA «CONECTA CON EL APRENDIZAJE Y CRECE»

Este año escolar 2024-2025 hemos establecido como lema «Conecta con el Aprendizaje y CRECE». Bajo este, damos a conocer nuevas herramientas curriculares diseñadas específicamente para mejorar la experiencia educativa de los estudiantes, los maestros y las familias. Estas herramientas permiten un seguimiento más detallado de la progresión de los aprendizajes, lo cual es clave para garantizar que los estudiantes avancen de manera constante y adecuada a su ritmo individual. Al ofrecer a los docentes una visión clara del progreso de cada estudiante y de cómo apoyarlos socioemocionalmente, se pueden identificar rápidamente aquellas áreas que necesitan refuerzo, así como también destacar los logros y habilidades adquiridas, individualizando la enseñanza para cada niño y joven.

Un aspecto fundamental para que nuestros niños y jóvenes *conecten con el aprendizaje y crezcan* es poner en práctica la enseñanza que sigue la progresión de aprendizaje, según los calendarios de secuencia. Por esto, las nuevas herramientas están alineadas al Currículo Oficial 2022 y diseñadas para facilitar un seguimiento detallado de la progresión de aprendizajes, lo que permite a los docentes identificar tanto los avances como las áreas que requieren apoyo adicional en cada estudiante. Estas herramientas también ofrecen recursos para contextualizar y diferenciar la enseñanza, adaptar los contenidos a las necesidades y a las características individuales de los estudiantes, para así garantizar una educación más personalizada y efectiva, enfocada en maximizar el potencial de cada uno.

Otro aspecto fundamental para que nuestros niños y jóvenes *conecten con el aprendizaje y crezcan* es el involucramiento de las familias en el aprendizaje de sus hijos. Cuando los padres están informados y comprometidos con el proceso educativo, pueden ofrecer el apoyo necesario en casa, refuerzan los aprendizajes y contribuyen activamente al éxito académico y al desarrollo socioemocional. Además, las nuevas herramientas proporcionan a las familias recursos y estrategias sencillas que les permiten participar de manera efectiva en el aprendizaje de sus hijos, promoviendo la creación de un entorno en el que el aprendizaje se extiende y se enriquece en el hogar.

Invitamos a todos a hacer suyo el lema «Conecta con el Aprendizaje y CRECE», mediante el uso de las nuevas herramientas curriculares diseñadas para mejorar la experiencia educativa de los niños y los jóvenes. Continuemos trabajando en fomentar el desarrollo holístico e integral de nuestros estudiantes. Aseguremos que tanto en la escuela como en el hogar se les proporcionen las herramientas y el acompañamiento necesarios para crecer como individuos completos. Esto impulsará su éxito académico y brindará una base sólida para el futuro personal y profesional de los niños y jóvenes ¡Conectémonos con el aprendizaje y crezcamos juntos!

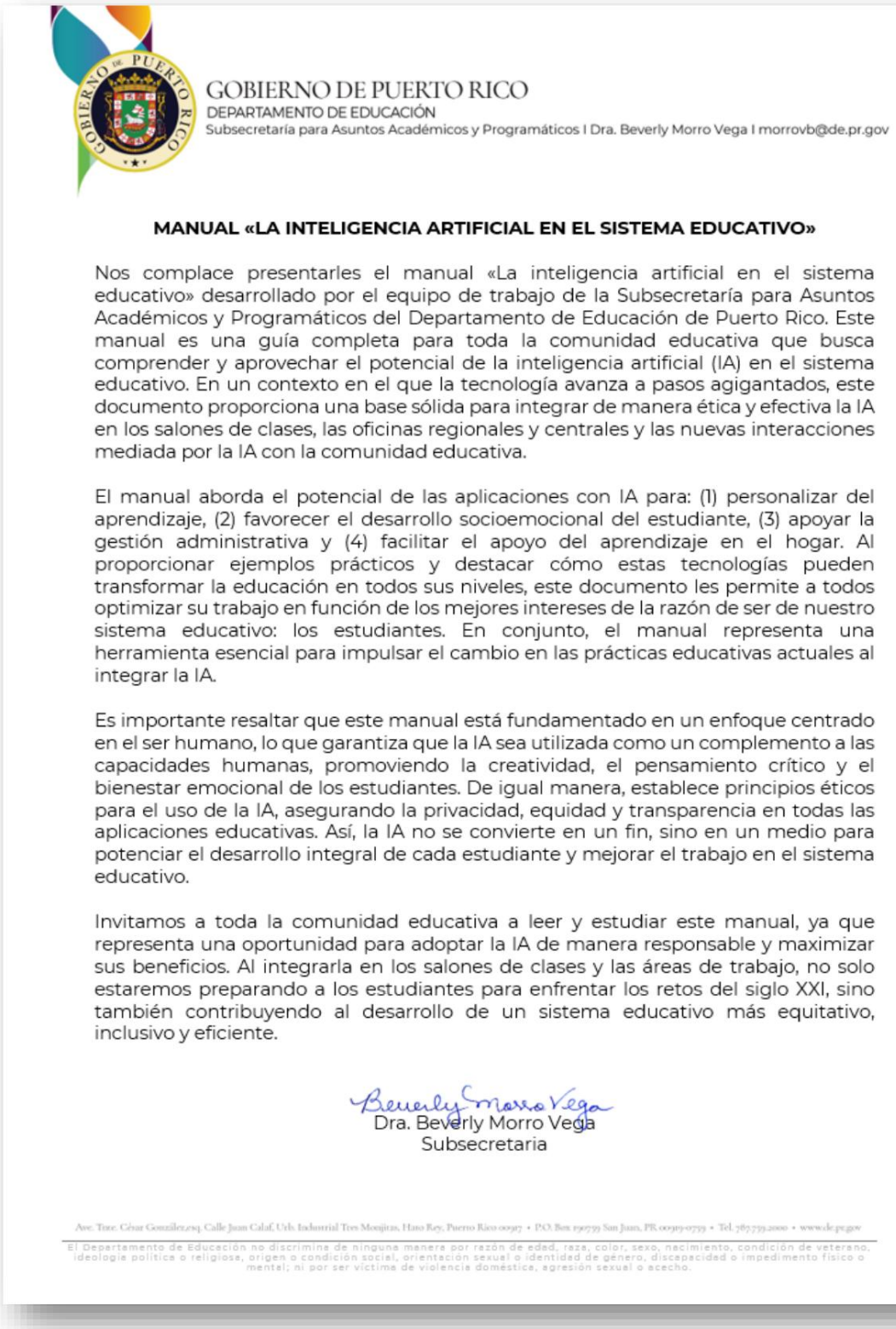
Dra. Yanira I. Raíces Vega
Secretaria

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MESSAGE FROM THE UNDERSECRETARY



GOBIERNO DE PUERTO RICO
DEPARTAMENTO DE EDUCACIÓN
Subsecretaría para Asuntos Académicos y Programáticos I Dra. Beverly Morro Vega I morrovb@de.pr.gov

MANUAL «LA INTELIGENCIA ARTIFICIAL EN EL SISTEMA EDUCATIVO»

Nos complace presentarles el manual «La inteligencia artificial en el sistema educativo» desarrollado por el equipo de trabajo de la Subsecretaría para Asuntos Académicos y Programáticos del Departamento de Educación de Puerto Rico. Este manual es una guía completa para toda la comunidad educativa que busca comprender y aprovechar el potencial de la inteligencia artificial (IA) en el sistema educativo. En un contexto en el que la tecnología avanza a pasos agigantados, este documento proporciona una base sólida para integrar de manera ética y efectiva la IA en los salones de clases, las oficinas regionales y centrales y las nuevas interacciones mediada por la IA con la comunidad educativa.

El manual aborda el potencial de las aplicaciones con IA para: (1) personalizar del aprendizaje, (2) favorecer el desarrollo socioemocional del estudiante, (3) apoyar la gestión administrativa y (4) facilitar el apoyo del aprendizaje en el hogar. Al proporcionar ejemplos prácticos y destacar cómo estas tecnologías pueden transformar la educación en todos sus niveles, este documento les permite a todos optimizar su trabajo en función de los mejores intereses de la razón de ser de nuestro sistema educativo: los estudiantes. En conjunto, el manual representa una herramienta esencial para impulsar el cambio en las prácticas educativas actuales al integrar la IA.

Es importante resaltar que este manual está fundamentado en un enfoque centrado en el ser humano, lo que garantiza que la IA sea utilizada como un complemento a las capacidades humanas, promoviendo la creatividad, el pensamiento crítico y el bienestar emocional de los estudiantes. De igual manera, establece principios éticos para el uso de la IA, asegurando la privacidad, equidad y transparencia en todas las aplicaciones educativas. Así, la IA no se convierte en un fin, sino en un medio para potenciar el desarrollo integral de cada estudiante y mejorar el trabajo en el sistema educativo.

Invitamos a toda la comunidad educativa a leer y estudiar este manual, ya que representa una oportunidad para adoptar la IA de manera responsable y maximizar sus beneficios. Al integrarla en los salones de clases y las áreas de trabajo, no solo estaremos preparando a los estudiantes para enfrentar los retos del siglo XXI, sino también contribuyendo al desarrollo de un sistema educativo más equitativo, inclusivo y eficiente.

Beverly Morro Vega
Dra. Beverly Morro Vega
Subsecretaria

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INTRODUCTION

The Artificial Intelligence (AI) revolution is rapidly changing various sectors of society, and education is no exception. It has quickly become an inevitable element of the dynamic teaching landscape at the primary, secondary and post-secondary levels. These new circumstances invite us to think carefully about the potential and scope of integrating AI into our educational system, as well as the possible challenges to achieve it and the expected results.

The integration of AI technologies into our educational model will focus on leveraging its capabilities to personalize learning and address individual academic needs to improve student academic achievement.

These qualities of AI technologies will allow us to expand our toolbox and adapt our teaching or work methodologies to prepare new generations for the challenges of the 21st century, while emphasizing their emotional and psychological development. This human component is crucial and makes a significant difference in ensuring that children and young people in Puerto Rico have "access to a liberating, comprehensive, and relevant education that will help them develop fully in their lives" (Law 85-2018, as amended).

From this perspective, the integration of AI technologies in the teaching and learning process – or at work – will allow the public education system to benefit from more flexible models that respond quickly to changes and needs in the environment in a timely and relevant manner. It will also be a transformative opportunity to foster closer collaboration between all actors in the educational scenario (students, families, teachers and administrators) and achieve more efficient and effective performance, according to each role.

However, it is essential to balance this flexibility with quality and ethics in teaching or work. That is why this document is a prescriptive one that aims to provide timely knowledge about AI technologies in learning and provide a frame of reference to determine how the various technological proposals with AI can be adapted and benefit our school or work communities. As all of us (teachers and non-teachers) in the Department of Education of Puerto Rico (DEPR) embrace the potential of AI technologies to revolutionize teaching and learning or work, we will have to face the harsh reality of inequality in access and the need for ethical considerations around its application.

It will therefore be essential to be open to the changes that AI technologies impose on the teaching and learning process or on working with critical thinking. This will be done in order to analyze each proposal, resource or strategy to ensure that AI technologies and the DEPR staff work in synergy to improve the educational process.

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LEGAL BASIS

Law 85 -2018, as amended, known as the "Puerto Rico Education Reform Act" established in article 1.02 - Declaration of public policy that:

d.1. The student is the sole reason for the existence of the educational system and the teacher is its main resource. All regulations must advance this goal and must not in any way hinder it. The secretary must thus state this in the regulations and circular letters. […].

d.3. The overall goal of education is to develop students to the fullest extent of their abilities and to ensure that they graduate prepared in STEM or STEAM subjects, so that they can compete in the global economy. At the same time, they should be allowed to develop their full personality and strengthen their appreciation for human rights.

d.4. Students will be educated in a holistic manner, taking into account their interests and ensuring that their particular needs are met. This includes ensuring their physical, emotional and mental well-being. […].

d.10. The student to whom the Department aspires is one with an integrated, holistic personality, sensitive as a citizen, committed to the common good, and with the skills and attitudes that allow him to contribute to Puerto Rico and his community in a proactive manner.

e. The educational management of the school must comply with the purposes that the Constitution and this Law establish for the Public Education System of Puerto Rico […].

The Elementary and Secondary Education Act of 1965 (ESEA), as amended by the Every Student Succeeds Act (ESSA), seeks to ensure that all students have a fair, equal, and significant opportunity to obtain a high-quality education and achieve, at a minimum, *proficiency* in the core subjects, according to the content standards established by the state. In addition, it enables the development of the Schoolwide project *in* schools. All processes carried out in schools, including school organization, will comply with the requirements established in this law.

The Carl D. Perkins V, *Career and Technical Education Improvement Act of* 2006, as amended, provides funding and regulations for the design and development of occupational and technical programs. The purpose of this act is to broadly develop the academic, occupational, and technical skills of secondary students enrolled in schools that offer these programs.

Act 51-1996, as amended, known as the " *Individuals with Disabilities Education Improvement Act* "
(IDEA) of 2004 and the Rehabilitation Act of 1973, as amended, set forth regulations for services provided to students with disabilities. Services must be consistent with the provisions of these, as applicable.

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Act No. 2 of 2019, as amended, known as the “Educational Continuity Act in the Event of Force Majeure Emergencies in the Public Education System of Puerto Rico,” provides for the creation of the process to be followed to avoid the interruption of the learning process of our students in cases of emergencies declared as such by the Governor of Puerto Rico and which cause the suspension of classes in the system; and for other related purposes.

Regulation No. 9193 of July 9, 2020, known as the "Regulations of Elementary and Secondary Schools of the Department of Education of Puerto Rico", contains the regulations on: admission of students, school organization, learning assessment, grade registration and grading system, study plan and promotion requirements, honor roll celebration, achievement and graduation day, teaching and learning planning, and School Excellence Design (DEE) in regular and occupational schools.

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FUNDAMENTALS OF DEPR

VISION AND MISSION

Vision

The Puerto Rico Department of Education aspires to be an effective instrument for the construction of a just and democratic society, cultivating ethics, solidarity and social awareness. It also aims to be a dynamic unit of social change, capable of explicitly developing the attitudes, skills and knowledge that prepare students competently and creatively to face the challenges of the modern world. It aims to respond to the varied needs and talents of students, diversifying offerings with creative learning and assessment alternatives, both during regular and extended hours.

Recognize the importance of developing knowledge and academic skills, in harmony with the emotional and social development of the student.

Mission

To ensure free, non-sectarian education that develops the attitudes, skills and knowledge of all students to function successfully in a globalized labor market and competently and creatively in accordance with the challenges of the modern world, as independent, lifelong learners, respectful of the law and the natural environment and capable of contributing to the common good.

PROGRAMMATIC GOALS

GOAL 1.	Improve the academic achievement of all students.
GOAL 2.	Integrate the participation of parents or guardians in the student's holistic education process.
GOAL 3.	Empowering school principals and teachers to make meaningful changes in their communities.
GOAL 4.	Strengthen technological infrastructure to ensure the teaching-learning process and data collection.

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GOAL 5.	Achieve administrative quality through decentralization of the system, the use of data for decision-making, effective use of the budget and a dynamic and efficient management system that manages operational, academic and fiscal processes in which performance, effectiveness and satisfaction indicators are used.
GOAL 6.	Promote physical, mental, emotional and social health through the services of the biopsychosocial model.

APPROACH TO AI INTEGRATION

The DEPR approach to integrating AI technologies is based on using technology as a tool that enhances, but does not replace, human intelligence and creativity, therefore, it is human-centered (Washington OSPI, 2024a [1]). This approach puts students, teachers and educational staff at the centre of the process, ensuring that AI serves to personalise teaching, improve well-being and support learning, always respecting the dignity, privacy and autonomy of each individual
(Valenzuela, S.; Contreras, A. & Rivera E.; 20242). AI should be used to enhance human capabilities, offering feedback, resources and analysis that empower educators to make informed decisions and help each student develop their full potential. In addition, it is essential that AI-based decisions are always accompanied by human reflection and judgment, promoting ethical, inclusive and equitable use that strengthens the well-being and integral development of all members of the educational community.
Washington OSPI, 2024a).

PURPOSE OF AI INTEGRATION WITH HUMAN-CENTERED APPROACH

The purpose of integrating artificial intelligence (AI) with a human-centered approach into the education system is to enhance the capabilities of students, teachers, and other employees to promote a more inclusive, equitable, and personalized education. Likewise, the goal is to use AI as a tool that supports, rather than replaces, creativity, critical thinking, and human judgment, allowing everyone who works at the DEPR to contribute, from each role, to optimize educational services so that students reach their full potential.

1 https://ospi.k12.wa.us/sites/default/files/2024-07/comprehensive-ai-guidance-accessible-format_0.pdf
2 https://repository.uaeh.edu.mx/revistas/index.php/sahagun/article/view/12574

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This approach also ensures that AI is used ethically and safely, respecting the privacy and well-being of all members of the educational community to ensure that technology is used to improve learning experiences and prevent inequalities or technological dependency. Ultimately, the intention is for AI to serve to enrich the educational process and prepare students to face a future in which they can collaborate with technologies in a critical and responsible manner.

AI INTEGRATION OBJECTIVES WITH A HUMAN-CENTERED APPROACH

1. Raise awareness about the use of AI in the entire school community.

2. Provide AI technology literacy to students and employees so that they understand the concepts, applications, and implications in various areas of knowledge.

3. Promote the norms and values of digital citizenship and responsibilities inherent to this.

4. Respect privacy and security by establishing and complying with ethical and fair standards that avoid bias and discrimination.

5. Personalize learning by adapting educational experiences to needs, individual skills and pace of each student or employee.

6. Provide staff, according to their role, with tools that allow them to improve their performance, optimize management in their workspace and make informed decisions based on data, without losing sight of human judgment.

7. Develop critical and creative problem-solving skills using AI to support exploring new ways of learning and thinking, without being fully dependent on technology.

8. Develop soft skills *as* key elements for collaboration, teamwork, academic and work performance, as well as well-being.

9. Promote responsible use of AI technology.

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ARTIFICIAL INTELLIGENCE

What is AI?

Artificial Intelligence (AI) is a branch of computer science that is dedicated to creating systems and machines capable of performing tasks that normally require human intelligence (AASA, 20233). These tasks include pattern recognition, decision making, language understanding, learning, and problem solving. Through algorithms and models, AI imitates or surpasses human capabilities in certain functions, allowing it to adapt and improve its performance based on the data with which it is trained. There are different types of AI, from reactive and predictive systems, to the most advanced ones such as generative AI, which can create original content from learned patterns (AASA, 2023 and Government of the Canary Islands, 20244).

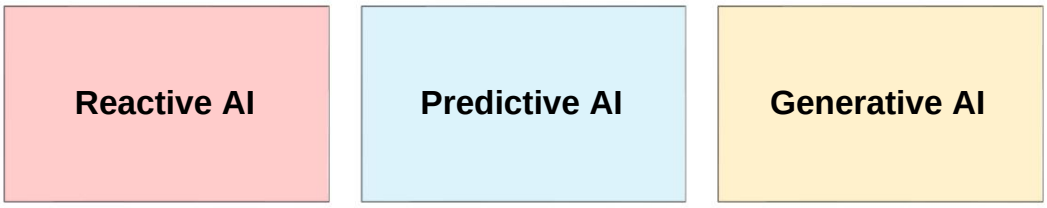
AI is used to perform tasks such as understanding language, recognizing patterns, or making decisions.

AI ranges from systems that perform tasks like recommending movies or auto-correcting typos, to more advanced forms that can generate new content or predict future outcomes.

AI has achieved significant development in recent decades, impacting multiple areas, from education to medicine and industry.
In education, for example, AI facilitates personalized learning, enables the creation of intelligent tutoring environments, and improves efficiency in school administration (Government of the Canary Islands, 2024). Despite its potential, its implementation entails ethical and social challenges, such as data privacy and equity in access to technology, so its responsible and safe use is crucial to maximize its benefits.

What are the types of AI?

According to the briefing paper *Bringing AI to School: Tips for School Leaders* (AASA, 2023), there are types of AI:



3 https://www.aasa.org/docs/default-source/advocacy/bringing-ai-to-school-tips-for-leaders.pdf
4https://www.gobiernodecanarias.org/cmsweb/export/sites/educacion/web/_galerias/descargas/otros/2024_02_29_guia_ia_en_educacion.pdf

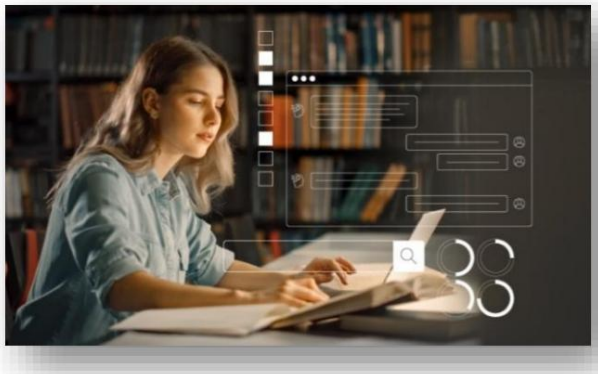
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1. **Reactive AI-** It is a type of artificial intelligence designed to respond to specific stimuli or situations without the ability to learn or store past experiences.
   These machines operate in real time and make decisions based solely on the information they have at the moment, without making future predictions or adaptations.
   A common example of reactive AI is virtual assistants such as Alexa or robot vacuum cleaners, which execute programmed actions based on received commands or detected conditions, but cannot improve or adjust their performance over time (AASA, 2023).

2. **Predictive AI -** This is a type of artificial intelligence that uses historical data and previous patterns to anticipate future events or behaviors. It works through algorithms that analyze large volumes of information to identify trends that allow for highly accurate predictions. This type of AI is common in systems such as content "recommenders" on *streaming* platforms or risk assessments in the financial sector, where it is anticipated which content, products, or services might interest a user based on their previous behavior. Predictive AI is widely used in sectors such as *marketing,* medicine, and education to optimize decisions and personalize experiences (AASA, 2023 and USDE Office of Educational Technology, 20235).

3. **Generative AI-** It is an advanced type of artificial intelligence that creates new content, such as text, images, music, or videos, from the patterns it has learned from large volumes of data. Unlike other types of AI that only analyze or predict, generative AI is capable of generating original creative products based on the data it has been trained with. An example of generative AI is models such as ChatGPT or MS Copilot, which can write coherent texts in response to questions, or tools such as DALL-E, which create images from textual descriptions. This technology has revolutionized fields such as digital art, content creation, and education, although it poses challenges regarding intellectual property and

   the ethical control of its use (AASA, 2023; UNESCO, 20246 and García-Peñalvo, FJ, 20247).



Imgane at https:// www.ebsco.com/ es/ tecnologia/ inteligencia-artificial-ai

5 https://tech.ed.gov/files/2023/05/ai-future-of-teaching-and-learning-report.pdf
6 https://unesdoc.unesco.org/ark:/48223/pf0000389227
7 https://www.researchgate.net/profile/Francisco-Garcia-Penalvo

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AI APPLICATIONS IN THE EDUCATION SYSTEM

Artificial intelligence (AI) applications are profoundly transforming the way education is taught, learned, and administered. It impacts the classroom, the workplace, the homes of students and employees, and life in the community (Rosario-Fernández, M., 20238 and USDE Office of Educational Technology, 2023).

1. **Classroom applications**

AI applications in the classroom focus on personalizing learning and addressing individual student needs, especially to improve the academic performance of all students, including those with special educational needs. First, AI enables the creation of platforms and tools that adapt educational content to the student's pace and abilities, adjusting difficulty levels and providing personalized feedback in real time. This is critical to delivering more inclusive teaching, as intelligent tutoring systems can identify areas where a child or young person needs additional support and suggest specific activities to reinforce their learning (Jara, I. & Ochoa JM, 20209 and USDE Office of Educational Technology, 2023).



Image at https:// queondagye.com/ artificial-intelligence-opportunity-or-threat-in-education/

In the case of students with special needs, AI also plays a crucial role in facilitating access to educational resources.

The AI facilitates the personalized of the learning tools through that adapt the contents to the individual needs of each student.

adaptive. Tools such as virtual assistants and voice recognition technologies enable these students to interact with content in ways that align with their abilities and learning styles.

For example, AI systems can assist students with hearing or visual impairments by providing automatic captions or descriptions of multimedia content. Additionally, AI platforms

8 https://www.researchgate.net/publication/373392496_La_Inteligencia_Artificial_en_Educacion_Hacia_un_Futuro_de_Aprendizaje_Inteligente
9 https://publications.iadb.org/es/publications/spanish/viewer/Usos-y-efectos-de-la-inteligencia-artificial-en-educacion.pdf

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They can generate individualized study plans, monitor each student's progress, and adjust teaching strategies according to their specific needs. This contributes to significantly improving their academic performance and closing learning gaps (Rosario-Fernández, M., 2023 and AASA, 2023).

2. **Applications in administrative offices**



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The applications of AI in the administrative field of schools and central offices are diverse and allow for the optimization of management and decision-making processes for superintendents, principals, and teacher facilitators. First, AI facilitates the automation of routine tasks such as school data management, processing of student records, resource allocation, and schedule planning, which frees up time and reduces the margin of error (Jara, I. & Ochoa JM, 2020). These tools also allow for the analysis of large volumes of data in real time, providing – to managers, superintendents, principals, facilitators, and even teachers – accurate information for making informed decisions, from school performance to the efficient use of financial and human resources (USDE Office of Educational Technology, 2023).

In regional educational offices (REOs) and central offices, AI can be used to improve efficiency in managing and overseeing multiple schools. For example, AI-based systems can automatically generate reports on key performance indicators, allowing educational leaders to spot trends and problems before they escalate (USDE Office of Educational Technology, 2023). It can also be used to optimize staff deployment, identify areas where more teacher support is needed, and manage teacher professional development through predictive analytics that identify teacher training needs.

Together, these applications help improve strategic decision-making, reduce operational costs, and ensure better coordination between the different areas of an educational organization (Washington OSPI, 2024a).

In this context, with the arrival of generative AI tools, the way in which administrative professionals (chief executives, operations managers, superintendents, school principals and teaching facilitators, administrative assistants, secretaries and office workers, among others) work is being significantly transformed.

AI optimizes school management by automating processes that allow for more efficient operation.

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work in offices. One of the biggest changes is the time saved on repetitive or administrative tasks, which allows staff to spend more time on strategic and high-value activities. With tools such as *MS Copilot,* among others, processes can be automated, for example: writing emails, preparing presentations and searching for relevant information in large amounts of data.

This improves productivity by reducing the time spent on data collection and analysis tasks, allowing consultants to make faster and better-informed decisions.

In addition, these generative AI tools are facilitating collaboration and communication between teams or with clients (students and their families, administrative, teaching and non-teaching staff). By generating visual and textual content more quickly, staff can present ideas and results more clearly and efficiently. This not only streamlines internal processes, but also improves the experience of the staff or individual who comes to seek a service, as

receive answers quickly and accurately. In the long term, the integration of AI in offices will allow employees to reinvent their roles, focusing on more creative and innovative tasks, while AI tools take care of the more mechanical or data-driven activities.



Image at https:// tynmagazine.com/ artificial-intelligence-how-positive-is-it-that-AI-comes-to-our-life/

(Microsoft, 202410).

3. **Home and community applications**

AI applications at home for students and their families are designed to facilitate learning, improve time management, and offer personalized support.
At home, students can use virtual assistants, such as *Alexa* or *Google Assistant,* to perform quick queries, organize study schedules, homework reminders, and receive immediate answers to school-related questions.

AI supports students in their tasks and studies through virtual assistants and adaptive platforms, facilitating autonomous learning.

These tools allow students to manage their time more efficiently and reinforce their learning outside of the classroom (USDE Office of Educational Technology, 2023).

There are also AI-based educational platforms that offer personalized learning experiences in

10 https://www.microsoft.com/en-us/worklab/how-business-leaders-are-using-copilot-right-now

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depending on the needs and progress of each student. Examples of these are:
that provide virtual tutoring and adaptive learning systems11 , which
They personalize educational content, based on the needs and pace of each student, which is useful for those with special educational needs or who require additional reinforcement. These tools allow children and young people to receive specific lessons and activities, helping them to improve in areas where they are struggling, while parents can monitor their progress and receive personalized recommendations on how to support learning at home (Rosario-Fernández, M., 2023; AASA, 2023 and Grezan, 202412).

On the other hand, families can use virtual assistants to help their children organize their study time, answer quick questions, and set reminders for assignments or exams. They can also turn to AI-powered virtual tutoring, which offers detailed explanations and real-time feedback to help them improve their learning.

facilitate learning in complex subjects. AI tools that generate and summarize content also help students establish an outline for a course.
Essays or research can be a useful resource for parents when they need to better understand a topic in order to support them. In this way, AI not only optimizes students' study time, but also facilitates the accompaniment of families in the educational process.



Image at https:// www.aarp.org/ espanol/ hogar-familia/ tecnologia/ info-2023/ educar-hijos-nietos-inteligencia-artificial.html

11 Adaptive learning is an educational approach that uses advanced technologies, such as artificial intelligence, to personalize the teaching and learning process based on each student's needs, abilities, and progress. Through digital platforms and systems, adaptive learning dynamically adjusts content, activities, and pace of instruction based on how each student interacts with the material and their performance in real time (AASA, 2023; and USDE Office of Educational Technology, 2023).
12 https://grezan.cl/la-inteligencia-artificial-en-la-educacion-latinoamericana-un-llamado-a-la-reflexion-y-colaboracion/

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ADVANCES AND BENEFITS OF AI IN EDUCATION

AI has emerged as one of the most transformative technologies in education, offering a wide range of applications that are revolutionizing the way education systems are taught, learned, and managed. From personalizing learning to automating administrative tasks, AI is enabling a

more efficient, equitable and accessible education for all. The advances and benefits provided by AI not only improve the educational experience of students, but also optimize educational processes in general, offering new opportunities for all members of the educational system.

ADVANCES AND BENEFITS	DESCRIPTION
a. Personalization of the learning	AI tools, such as adaptive learning platforms, adapt content and activities to the individual needs of each student, offering personalized learning according to their pace and learning style (UNESCO, 202313 and INTEF, 202414).
b. Improving academic performance	AI-based platforms or systems provide immediate feedback, helping students, teachers, and families to quickly identify and correct errors, which in turn contributes to improving their performance (INTEF, 2024).
c. Equity and inclusion for students with functional diversity	AI facilitates access to adapted resources for students with special needs in order to reduce barriers and promote more inclusive education (UNESCO, 2023).
d. Automation of administrative tasks	Various AI systems allow for the automation of processes such as grade management, resource allocation and schedule organization, content adaptation or execution tasks, which frees up time for pedagogical tasks and reduces the administrative burden (Oliver, N., 202015 and UNESCO & Fundación SM, 202216).
e. Evaluation of the efficient and personalized learning	AI-based platforms or systems enable continuous and detailed evaluation of the progress of the

13 https://unesdoc.unesco.org/ark:/48223/pf0000388894
14 https://code.intef.es/wp-content/uploads/2024/07/Gu%C3%ADa-sobre-el-uso-de-la-IA-en-el-%C3%A1mbito-educativo- INTEF_2024.pdf
15 https://portal.mineco.gob.es/RecursosProducto/mineco/ministerio/ficheros/libreria/Inteligencia_artificial_naturalmente.pdf
16 https://unesdoc.unesco.org/ark:/48223/pf0000381560

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ADVANCES AND BENEFITS	DESCRIPTION
	students by providing real-time performance analysis and improvement recommendations (INTEF, 2024).
f. Creation and adaptation of content	AI tools help create personalized educational resources, adapted to the characteristics and levels of students; this offers a better learning experience (UNESCO, 202117).
g. Access to resources digital educational	AI technologies facilitate access to online educational platforms that offer automated lessons, exercises and tutorials, allowing learning anytime, anywhere (UNESCO & Fundación SM, 2022).
h. Support for data-driven educational decision-making	Data generated by AI systems allows teachers and administrators to make informed decisions about teaching strategies to improve educational planning and management (UNESCO, 2023 and UNESCO & SM Foundation, 2022).
i. Promotion of critical and creative thinking	AI generates learning scenarios that stimulate the development of critical and creative thinking by presenting students with complex problems that they must solve (INTEF, 2024).
j. Training and professional development for teaching staff	AI is applied in teacher training, offering personalized training and pedagogical support resources to improve their skills and effectiveness in the classroom (INTEF, 2024 and UNESCO & Fundación SM, 2022).
k. Development of soft *skills*	AI contributes to the development of soft skills such as communication and teamwork through digital collaborative environments (Cardon, P., 202418)

17 https://unesdoc.unesco.org/ark:/48223/pf0000381137_spa
18 How AI is making soft skills more important in the workplace | World Economic Forum (weforum.org)

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CHALLENGES AND RISKS OF AI IN THE EDUCATION SYSTEM

The integration of AI into the education system presents great opportunities, but also brings with it a number of challenges and risks that must be carefully addressed to ensure effective, efficient, and ethical use. As AI is increasingly adopted in classrooms and school management, concerns arise about equity in access, privacy and security of student data, as well as the impact on the role of teachers and the development of critical skills in students. Therefore, below are the challenges and risks that we as an education system must consider when implementing AI in the classroom and in the workplace.

Challenges

According to the literature review, there are three types of challenges:

Ethical challenges	Challenges in data privacy and security	Challenges of dependency and student autonomy and staff
1. Equal access to AI technologies and connectivity- AI can exacerbate the existing inequalities if equitable access to technology is not guaranteed, especially communities in disadvantaged (UNESCO, 2023).	**1. Data protection Personal data of students and employees -** AI in education requires the use of large volumes of personal data, which poses significant risks if adequate protection against leaks or unauthorized access is not ensured. accesses No (UNESCO, 2022 & INTEF, 2024).	**1. Teacher professional development, training of families and students -** Lack of adequate training about AI among teachers and students and their families hinders its effective implementation in the classroom and at home (INTEF, 2024).
2. Teaching role - AI could reduce the human interaction in the education, which would affect the relationship between teachers and students. No However, he DEPR he considers the teacher as your main resource that facilitates learning, and will favor the relationship	**2. Consent to use of information-** It is necessary ensure that students and their families understand how their data is used and provide informed consent (UNESCO & Fundación SM, 2022). In addition, it requires that staff with access to personal data remember the	**2. Digital divide-** Inequalities in access to technology and connectivity limit the potential of AI to benefit all students, especially exacerbating educational inequalities between students. of different levels socioeconomic (UNESCO, 2023 & INTEF, 2024).

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Ethical challenges	Challenges in data privacy and security	Challenges of dependency and autonomy of students and staff
synergistic between the teacher and AI (UNESCO, 2022).	requirements of the FERPA19, COPPA20 and CIPA21 Act .	
3. Responsible use of data- Taking decisions AI-based decisions requires responsible data management to avoid vulnerability the rights of the students (UNESCO, 2022), so, outside of the DEPR network and the tools it provides, staff must use their criticality to determine the security of the AI technology to be used.		



Image at https:// news.microsoft.com/ en-us/ 7-ways-ai-can-take-business-management-and-production-to-the-next-level

Risks

According to the literature review, there are two types of risks:

Risks of algorithmic bias	Risk of dependency and autonomy due to over-reliance on AI
1. Discrimination and Exclusion- AI algorithms22 can incorporate biases present in the data they were trained on. This could lead to unfair decisions.	**1. Risk of AI dependency in the student-** The excessive use of AI tools AI to solve problems or complete tasks could limit skill development

[19] Family Educational Rights and Privacy Act (FERPA): https://studentprivacy.ed.gov/sites/default/files/resource_document/file/Ferpa-for-parents-spanish.pdf
20 Children's Online Privacy Protection Rule (COPPA): https://www.ftc.gov/legal-library/browse/rules/childrens-online-privacy-protection-rule-coppa
[21] . Children's Internet Protection Act (CIPA): https://www.fcc.gov/consumers/guides/childrens-internet-protection-act
22 «A machine learning algorithm is a set of rules or processes used by an AI system to perform tasks, most often to discover new insights and data patterns, or to predict output values from a given set of input variables. Algorithms enable machine learning (ML) to learn» [IBM, (n.d.). What is a machine learning algorithm? IBM. https://www.ibm.com/mx-es/topics/machine-learning-

algorithms#:~:text=Un%20algoritmo%20de%20aprendizaje%20autom%C3%A1tico%20es%20un%20conjunto%20de%20reglas,determinado%20de%20variabl[is%20input]

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Risks of algorithmic bias	Risk of dependency and autonomy due to over-reliance on AI
unfair or discriminatory, especially in the evaluation and recommendation of educational content, which is why the expert eye, that is, the educator, is key in their role as learning facilitator (INTEF, 2024 and UNESCO & Fundación SM, 2022).	fundamentals such as critical thinking, creativity and the ability to make decisions independently (INTEF, 2024). When students become too dependent on automated solutions, their initiative to face challenges on their own is reduced, and their ability to manage and direct their own learning process is limited.
2. Lack of transparency- Lack of transparency in artificial intelligence (AI) algorithms is one of the main risks in their integration into the education system. AI algorithms, which are used to personalize learning, evaluate student performance, and improve the quality of education, are not transparent students or automate tasks Administrative systems often operate as "black boxes"; that is, their internal processes are not fully understandable to end users, including teachers, administrators or students. This opacity makes it difficult to monitor and control how automated decisions are made and what criteria they are based on (INTEF, 2024).	**2. Risk of AI dependency on staff-** Educational staff are also at risk of over-reliance on AI for teaching, assessment and management functions, as it can diminish their performance in the educational process, depending on their role. Although AI can automate certain administrative tasks and provide data analysis, over-reliance could lead to a decrease in human intervention; therefore, would affect the quality of teaching and pedagogical decision-making (INTEF, 2024). This is why it is important to find a balance in the use of AI, to ensure that it is a complementary tool and not a substitute for the active participation of students and teachers or work teams.

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POSSIBLE USES OF AI IN THE EDUCATION SYSTEM

AI offers tools that facilitate the personalization of learning. Teachers can thus adapt their methodologies to the specific needs of each student, while principals and superintendents can optimize decision-making and school management through data analysis and process automation. Below are potential uses of AI in curriculum and educational operations that highlight its potential to improve academic performance, administrative efficiency, and professional development of teachers. We invite everyone to reflect on these recommendations and consider them when selecting or adopting AI tools for their professional practice in an ethical and safe manner. In this way, we ensure that the use of AI drives a more inclusive, effective, and human-centered education.

a. To integrate AI technologies into the curriculum and personalize learning

1. Teach students to understand the basic concepts of AI, its applications and limitations, encouraging critical and responsible use of these tools.
2. Use AI tools to generate problem-solving scenarios that promote the development of critical and creative thinking.
3. Use AI-powered technologies to design assessments that are tailored to each student's needs and competencies, while always complying with FERPA, to measure their individual progress more accurately.
4. Apply AI-powered technologies that help create differentiated activities for groups with different abilities to help students work at their own pace without overburdening those who need more time.
5. Use AI tools to create adapted resources that improve accessibility and the learning experience of students with special educational needs.

6. Implement AI tools that automatically adjust educational content *(adaptive learning)* based on the pace and level of each student.
7. Use AI technologies to offer personalized tutoring that reinforces areas where students need more support to close learning gaps.

8. Introduce virtual assistants or AI platforms that students can use use at home to reinforce their learning.
9. Integrate AI tools that allow immediate feedback to customers. students to reinforce their learning by being corrected quickly.

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10. Apply AI tools that promote collaboration or teamwork to encourage the development of soft skills .

b. To integrate AI technologies into teaching administrative work

1. Implement AI-powered tools that automate routine tasks like grading tests, tracking student progress, and generating reports.

2. Use AI-powered systems to create dynamic weekly plans that fit the needs of each group, prioritizing closing learning gaps.
3. Employ AI to collect and analyze academic performance data quickly and effectively to make informed academic decisions to improve student learning.

4. Apply predictive analytics AI tools that help you identify patterns in student performance and anticipate potential risk areas
to implement more effective early intervention strategies.

c. To integrate AI technologies into the integrated support process

1. Use AI-powered tools to automate history management, reporting, and
student records, always in compliance with FERPA.
2. Implement AI platforms that allow continuous monitoring of student progress in different areas.

3. Use virtual assistants or AI tools to manage automatic reminders for meetings with students, parents or school staff, to improve efficiency in managing appointments and follow-ups.

4. Use AI resources to generate personalized reports on the emotional, social or health status of students, based on data collected during interactions, always complying with FERPA.

5. Integrate AI systems that identify patterns in student behavior, attendance, or academic performance, helping to detect early signs of risk of dropping out of school, emotional problems, or academic difficulties to provide timely follow-up.

6. Use AI tools that issue automatic alerts when behaviors suggesting risk of depression, anxiety or other emotional problems are detected.

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7. Implement AI-based questionnaires and assessments to monitor students' emotional well-being on a regular basis, generating reports that allow intervention in case of detecting situations of stress or discomfort.
8. Use AI tools that automatically manage follow-up plans for students with special needs to ensure that intervention and care protocols are followed.

9. Enter:
    a) Virtual assistants who can interact with students in times of stress or crisis to provide immediate emotional support.

    b) AI-powered tools that allow students to conduct regular self-assessments of their emotional well-being to facilitate the identification and communication of their support needs in time.

10. Integrate platforms that use AI to mediate conflicts between students, providing impartial solutions and recommendations to reduce tensions and improve school coexistence.

11. Use AI systems to collaboratively create resources and guides to help manage emotional emergencies, assaults, or life-threatening events.
traumatic.
12. Apply AI tools that detect stress levels related to academic performance to carry out interventions at critical moments of the school year.

13. Use AI to design individualized support plans, both emotionally and physically, that include personalized recommendations on healthy activities, stress management, and self-care routines.

14. Introduce virtual assistants or AI-based platforms that help customers
students to:
    a) manage your study time, set reminders for tasks and organize your schedules.

    b) break down large tasks or projects into smaller, more manageable steps.
    c) improve self-organization, concentration and productivity in the process of studying.

d. To integrate AI technologies to improve administrative efficiency

1. Use AI tools to automate routine administrative tasks such as record management, scheduling, and report generation, among others.

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2. Implement AI-powered systems to analyze large volumes of data (attendance, academic performance, financial resources) and generate predictive analytics to support informed decision-making and identify trends and areas for improvement.

3. Use AI to optimize the allocation of physical, human or material resources.
4. Integrate AI-based platforms to manage recruitment processes, staff performance tracking or professional development.
5. Introduce AI enhancement to existing platforms to continuously track student attendance and academic performance, generating early alerts about absences or declines in performance in order to intervene in a timely manner.

6. Use AI-based virtual assistants or *chatbots*[23] to manage communication with parents, students, and staff, providing automated responses to frequently asked questions or reminders about important dates.
7. Use AI to manage educational data collection and analysis to create detailed reports and automatic visualizations that facilitate understanding of key results and support the development of school improvement plans.
8. Apply AI-powered tools that analyze spending patterns and recommend adjustments to school budget allocation to maximize efficiency and ensure resources are used optimally.

9. Introduce AI-enabled technologies to gather real-time feedback from staff, students, and families on administrative initiatives and processes to make quick adjustments and continuously improve operational effectiveness.



Image from https:// www.shutterstock.com/ es/

23 It is a computer program that simulates a conversation with human end users [IBM, (sf). What is a chatbot? IBM. https://www.ibm.com/es- es/topics/chatbots]

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ETHICAL PRINCIPLES FOR THE USE OF ARTIFICIAL INTELLIGENCE

BEGINNING	THE THEREFORE
1. Transparency and clarity	ÿ The implementation of AI at the DEPR must be completely transparent to students, teachers, families, and administrators. They must be informed about how AI is implemented. use AI tools, what data is collected and shared, how it is processed, for what purposes, and how the Family Educational Rights and Privacy Act (FERPA) and the Children's Online Privacy Protection Act (COPPA) are complied with. ÿ *Therefore, OREs and schools must provide clear and accessible documentation on the use of AI, ensuring that the entire educational community understands the benefits, risks and implications of using the technology.*
2. Equity and non-discrimination	ÿ AI should promote educational equity and not perpetuate existing biases that lead to discrimination. All DEPR students, regardless of their socioeconomic background, gender, ethnicity, abilities or disabilities, should have equal access to the benefits of AI. ÿ *AI technologies should therefore be regularly monitored to identify and mitigate potential algorithmic biases that may favour certain groups of students over others. In addition, OREs and schools should ensure that all students have equal access to the technology needed to participate in AI-powered environments.*
3. Privacy and data security	ÿ Protecting student data is a priority, so the PRDE must ensure that AI systems comply with all applicable data protection regulations, such as FERPA and COPPA. ÿ *Therefore, OREs and schools must implement clear policies on the collection, storage and use of student data, ensuring that these are only used for educational purposes and that they are managed securely and responsibly within the DEPR network.*
4. Human responsibility and supervision	ÿ AI should be used as a support tool for students, educators, and administrators, not as a replacement. Educational decision-making that impacts students should always be under the supervision and control of teachers and system administrators, who are the those responsible for decisions arising from AI systems.

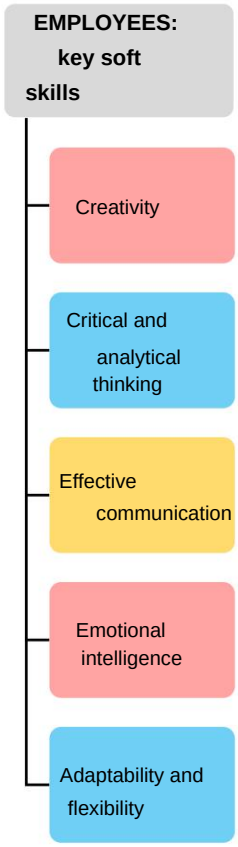
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BEGINNING	THE THEREFORE
	ÿ *Therefore, there must always be human intervention in decisions that affect the well-being of students.* *Consequently, the DEPR, through the OREs and schools, must ensure that teachers receive adequate training on how to use AI tools responsibly and how to interpret the results generated by AI technology.*
5. Autonomy of the teacher and student	ÿ While AI can provide personalization and support, it should not limit the autonomy of teachers to adjust their teaching methods or of students to actively participate in their own learning. ÿ *Therefore, AI should serve to enhance the capabilities of teachers and students, not to restrict them. Teachers, in accordance with the teacher autonomy granted and defined in article 2.16 of Law 85-2018, as amended, have the flexibility to adapt the results generated by AI to the particular needs and contexts of their classrooms, ensuring that AI technologies complement, but do not dominate, the teaching process.*
6. Charity and educational improvement	ÿ The implementation of AI should aim to improve student learning and well-being, contributing to their comprehensive development. Any use of AI in education should be geared towards maximizing the benefits for students and educational communities. ÿ *Therefore, before implementing AI technologies, OREs and schools must evaluate and prioritize the clear benefits that these tools can bring in terms of pedagogical efficiency, personalization of learning and reduction of inequalities.*
7. Evaluation continue	ÿ AI technologies should be monitored and evaluated on a regular basis to ensure that they continue to meet their intended educational and ethical objectives. ÿ *Therefore, the DEPR, through its Systems Office, must establish periodic monitoring and continuous evaluation mechanisms to ensure that the acquired AI systems or technologies remain effective and respect ethical principles. In addition, adjustments must be made based on the results obtained. Likewise, guidelines must be established on the technical qualities and considerations for access to information in these systems.*

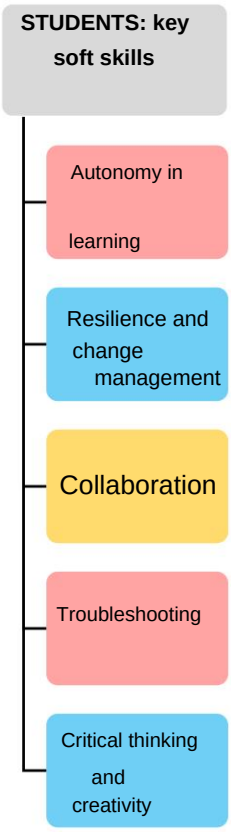
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DEVELOPING SOFT SKILLS INTEGRATING AI

The increasing integration of AI into the education system is transforming both the way students learn and the way teachers and other employees manage their duties and responsibilities. While AI offers countless benefits, such as personalizing learning and streamlining administrative tasks, it also poses a fundamental challenge: the need to strengthen soft *skills.* These skills, which include creativity, critical thinking, emotional intelligence, and adaptability, are essential for both students and employees to make the most of technological tools without losing sight of their integral human development. In this context, developing these soft skills is not only a competitive advantage, but a crucial necessity in education and at work.



For employees in the education system, *soft skills* are essential to ensure effective implementation of AI in their daily work. Skills such as



Effective communication and emotional intelligence are vital to managing technological change without compromising human interaction in schools. Since AI can automate many routine tasks, it is important for employees to focus on areas that require human judgment,

empathy and proper emotional management (Spencer, J., 2024a & Skill Cycle, 2024). In addition, adaptability and flexibility allow employees to quickly adjust to new technological tools and processes, while maintaining a focus on student well-being and administrative efficiency (Marr, B., 2020).

For students, *soft skills* are equally important to prepare them for an increasingly automated environment. While AI can facilitate access to information and personalized learning, students must develop skills such as autonomy and critical thinking to be responsible for their own learning (Training Industry, 2023 & Skills You Need, 2024). Creativity and problem-solving are also key, as students are more likely to be able to use their skills to their advantage.

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Students must learn to question information generated by AI and use it to generate innovative solutions. Without these skills, students run the risk of becoming overly dependent on technology without developing the cognitive skills necessary to meet real-world challenges (Spencer, J., 2024a).

Another crucial aspect is the importance of collaboration in the context of AI. Even though technology enables individualized learning, teamwork skills remain critical for both employees and students. AI cannot replicate the human ability to collaborate effectively in diverse teams, and this is where *soft skills* such as communication and empathy play a central role (Spencer, J., 2024a World Economic Forum, 2024). The DEPR is an educational system that values these skills, therefore, employees and students can use AI technologies

as a complementary tool, without losing sight of the importance of the human interactions.

In conclusion, the integration of AI into the education system raises the need to develop and strengthen *soft skills* in both employees and students. These soft skills not only improve the effectiveness in using technology, but also ensure that the educational approach is human-centered. Critical thinking, creativity, collaboration, and emotional intelligence are skills that allow us to take full advantage of the benefits of AI, while preparing individuals for a future in which human competencies will be more valuable than ever. As we move towards a more digital education system, *soft skills* become the essential bridge between technology and comprehensive human development.

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GENERATIVE AI AND POWERFUL QUESTION ASKING

Generative AI, which includes models such as text generators and image creators, has shown immense potential to transform the education system. Its ability to produce original content, from detailed responses to interactive activities, allows students, teachers, and other staff to access unlimited resources that enhance learning, teaching, or work. However, to maximize its effective use in the educational environment, it is essential that all of them learn to ask powerful questions. Well-structured questions not only optimize the results generated by AI, but also promote deeper and more meaningful learning.

Asking powerful questions is a key skill to fully harness the potential of generative AI. Questions that are clear, specific, and geared toward exploring complex concepts, problem-solving, or critical analysis allow AI to offer richer, more detailed answers.

For example, instead of simply asking "What is climate change?" a powerful question would be "How does climate change affect ecosystems in different regions, and what measures can mitigate these effects?" The latter question not only elicits a more complex and nuanced response, but also invites the student to critically reflect on the topic.



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The strategic use of powerful questions also helps students develop important skills such as synthesis, critical analysis, and creativity.

When students learn to ask questions that explore multiple perspectives or scenarios, they develop deeper thinking that is geared toward solving complex problems. Generative AI, then, becomes a valuable tool that not only offers information, but guides students in the process of learning how to learn, exploring ideas more autonomously and creatively.

Despite the advantages that generative AI offers in education, it is essential that learning does not depend exclusively on these technologies. Unmediated study

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Technology remains essential to the comprehensive development of all members of the education system. Formulating powerful questions should not be contingent on technology, but should foster a thinking process that students, teachers, and other staff can apply independently of AI. Learning or studying without technology strengthens everyone's ability to research, analyze, and reflect independently, skills that are fundamental to their academic and personal development.

In the context of teaching, asking powerful questions also enables educators to use generative AI to create personalized and adaptive resources. By asking detailed questions, teachers can come up with ideas for differentiated activities, personalized assessments, and support materials that fit the specific needs of each group of students. This not only maximizes the use of generative AI, but also enables more effective, student-centered teaching, promoting more equitable and personalized learning.

On the other hand, in the context of non-teaching work, asking powerful questions makes it possible to optimize the use of AI to improve administrative efficiency and strategic decision-making. By asking clear and well-structured questions, administrators and school leaders can obtain more accurate and relevant information on key aspects of resource management, staff distribution, student performance and, based on these, request predictive analysis. Moreover, asking powerful questions in this area could anticipate problems and solve them before they become major challenges.

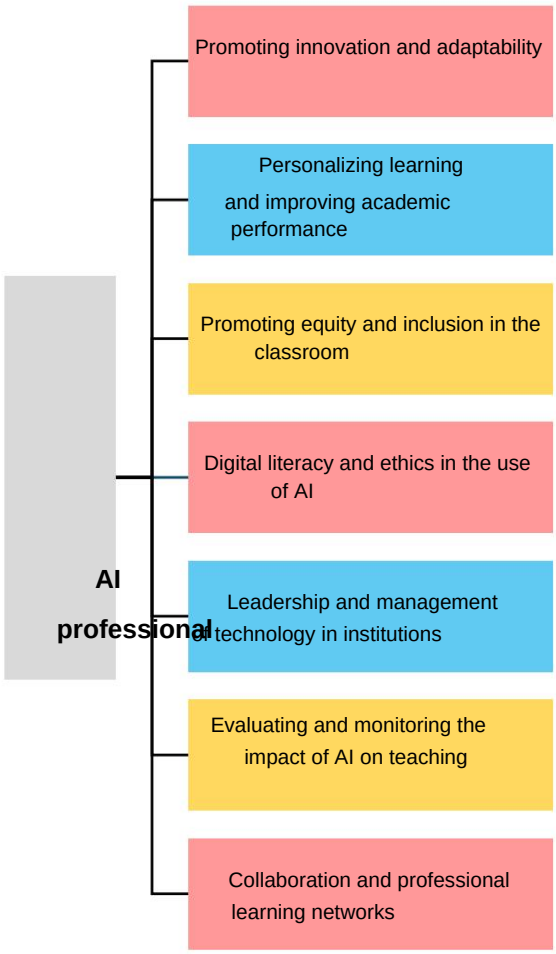
This type of powerful question-based approach maximizes the value of Generative AI technology tools and empowers everyone's capacity for critical and creative thinking and data-driven problem solving, while leveraging AI to improve communication and collaboration within the institution.



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THE IMPORTANCE OF PROFESSIONAL DEVELOPMENT IN THE USE OF AI



Professional development is critical to ensure that educators effectively integrate AI into their daily teaching practice.

AI offers the potential to transform teaching, but its success depends on being properly trained, not only in the technical use of the tools, but also in understanding how to apply them to enhance learning and meet the individual needs of students.

Below are key reasons why professional development is essential in integrating AI into education.

1. Fostering Innovation and Adaptability – As AI is a constantly evolving technology, in order for teachers to stay up to date and take full advantage of its benefits, it is crucial to offer ongoing professional development that promotes innovation. Teachers need to stay abreast of the latest developments, pedagogical strategies, and best practices to effectively integrate AI.

a. Core themes: pedagogical innovation, use of emerging technologies, continuous updating in AI applied to education.

2. Personalization of learning and improved academic performance – One of the biggest benefits of AI is its ability to personalize learning based on the needs and styles of each student. However, for this personalization to be effective, teachers must be trained in how to interpret the data generated by AI and how to adjust their teaching methods accordingly.

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   a. **Core topics:** personalization of learning, analysis of educational data, adjustment of pedagogical strategies based on AI information.

3. **Promoting equity and inclusion in the classroom-** AI has the potential to reduce inequalities in education by offering tools that support students with diverse abilities and socioeconomic backgrounds. However, it is critical for teachers to understand how to use AI inclusively and ensure that it is accessible to all.
   that all students, regardless of their abilities and resources, have access to the benefits that technology offers.

   a. **Core issues:** equity in access to technology, inclusive strategies, reducing learning gaps through AI, attention to students with special needs.

4. **Digital literacy and ethics in the use of AI –** It is essential that teachers not only know how to use AI, but also understand the ethical implications of its use. AI can raise questions about data privacy, human control, and fairness. Therefore, teachers must be trained to address these issues in an informed and ethical manner.

   a. **Core topics:** ethics in the use of AI, data protection and privacy, digital rights of students, evaluation of biases in algorithms.

5. **Leadership and management of technology in institutions –** In addition to training teachers, it is important that school administrators receive support to lead the implementation of AI in their institutions. Administrators must be able to create an enabling environment for the adoption of technologies, as well as ensure that teachers have the necessary support and that infrastructures are adequately designed to integrate AI.

   a. **Core issues:** technology management, AI implementation policies, educational leadership in the digital age.

6. **Evaluating and monitoring the impact of AI on teaching –** To ensure that AI integration is successful, it is essential that educators and administrators have the tools to monitor and evaluate its impact. This includes the ability to measure how AI tools are affecting student performance and make adjustments to instructional practices as needed.

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   a. **Core topics:** assessing the impact of AI, analyzing educational outcomes, continuous monitoring and adjustments in AI implementation.

7. **Collaboration and professional learning networks** – AI training should not be a one-off event, but rather a continuous learning process. Creating collaborative networks between teachers allows them to share their experiences and best practices in using AI. Teachers should have access to forums and discussion groups where they can can exchange ideas and learn from their colleagues.

   a. **Core themes:** professional learning networks, teacher collaboration, peer mentoring and support, soft skills development , and AI.

Professional development in the use of AI in education is essential for teachers and administrators to be able to take full advantage of the potential of these tools. This should be adapted to the specialty of each staff member, according to their position and responsibilities. Finally, we remind everyone that professional development begins with self-education. We invite everyone to review the reference list in this document and begin their study process.

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CONCLUSION

The continued evolution of AI promises to have an ever-deepening impact on the education system, opening up new opportunities to personalize learning, optimize processes, and improve academic outcomes. As the technology advances, we are likely to see greater integration of generative AI, adaptive learning, and predictive analytics tools into classrooms and administrative offices. These technologies will enable educators to deliver more dynamic and personalized learning experiences, while administrators will be able to make more informed and efficient decisions. However, this evolution must be accompanied by a firm commitment to ethical and human-centered use.

A human-centred approach ensures that AI is used as a tool to empower, not to replace, the development of skills, attitudes or the execution of tasks. This means that technology must always support and respect human capabilities, while fostering critical thinking, creativity and autonomy. AI can facilitate teaching, but it is crucial that educators remain the primary guides of learning, using technology to complement, not displace, their pedagogical role. This balance will ensure that AI truly serves to enrich the educational process and not create an over-reliance on technology.

The future of AI in education also involves a collective responsibility to address ethical challenges such as equity in access, data privacy and transparency of algorithms. While AI has the potential to close educational gaps, there is a risk that it will widen inequalities if equitable access to the technology is not guaranteed. Furthermore, handling large volumes of students' personal data requires a rigorous approach to protecting privacy and security. Transparency in the functioning of algorithms will be essential to avoid bias and ensure that AI-based decisions are fair and equitable.

In this regard, it is vital that educational policies on AI are designed with a strong commitment to the common good. This means that all AI applications in the education system must focus on improving the educational experience of students and teachers, ensuring that their rights and dignity are respected. Data-driven decisions and the use of advanced technologies must be guided by principles of justice, equity and inclusion, so that every student or employee has the opportunity to benefit from technological advances, regardless of their socioeconomic background.

In conclusion, the evolution of AI in education has the potential to radically transform the educational system, by offering new tools to personalize the

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learning and improving administrative management. However, this development must be accompanied by an ethical and human-centred approach, which at the same time guarantees that technology serves to enhance human capabilities and promote the well-being of the entire educational community. Only in this way can we ensure that AI in education effectively contributes to the common good and to the construction of a more just and inclusive society.



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REFERENCES

Acar, OA (2024). With generative AI we can reimagine education — and the sky is the limit . *World Economic Forum.* https://en.weforum.org/agenda/2024/02/with-generative-ai-we-can-reimagine-education-and-the-sky-is-the-limit/

Alarcón-San Lucas, MV, & Millán-Ibarra, AI (2021). The relationship between the responsible use of technology and learning in students of the Basic Secondary Sublevel of the Leónidas Plaza School. *593 Digital Publisher CEIT, 6 (2-1), 120-138.* https://doi.org/10.33386/593dp.2021.2-1.528

Ayuso-del Puerto, D., & Gutiérrez-Esteban, P. (2022). Artificial intelligence as a resource educational during initial teacher training. *RIED. Ibero-American Journal of Distance Education, 25 (2), 347-368.* https://doi.org/10.5944/ried.25.2.32332

Cabrera, J. (2023, July 22). The impact of AI on the education of the future: A revealing conversation with ChatGPT. *The power of redarchy.* https://cabreramc.com/the-impact-of-ai-on-the-education-of-the-future-a-revealing-conversation-with-chapgpt/

Chicago Public Schools. (2024). *AI guidebook.* Chicago Public Schools. https://www.cps.edu/strategic-initiatives/IA-guidebook/

Codemotion. (2023). *What Soft Skills are Key in The Era of AI?* https://www.codemotion.com/magazine/IA-ml/what-soft-skills-are-key-in-the-era-of-IA/

Department of Education, Vocational Training, Physical Activity and Sports. (2024). *The Artificial intelligence in the educational field.* Government of the Canary Islands. https://www3.gobiernodecanarias.org/medusa/ecoescuela/ia/guia/

Espinosa García, P. (2023). Artificial intelligence and soft skills: Why are they important? important for work? *Excites.* https://emocional.com/artificial-intelligence-and-soft-skills-why-are-they-important-for-work

ElIA.io. (2024). AI *Evolution in Soft Skills TrAIning: Shaping the Future of Professional Development.* https://elIA.io/soft-skills-trIAning-with-IA

Farnós, JD (2024). Building a BASE MAP: University takeoff in the 21st century within the framework of disruptive education and artificial intelligence. *Juandon. Innovation and knowledge.* https://juandomingofarnos.wordpress.com/2024/09/02/construction-of-a-map-

Machine Translated by Google

base-takeoff-of-the-university-in-the-21st-century-within-the-framework-of- disruptive-education-artificial-intelligence/

Fast Company. (2024). *The human edge: Why soft skills matter more than ever in the AI era.* https://www.fastcompany.com/91178670/the-human-edge-why-soft-skills-matter-more-than-ever-in-the-IA-era

Fernández de Silva, M.R. (2023). *Artificial intelligence in education: Towards a future of intelligent learning.* Scribe. ISBN 978-980-7898-54-6. https://dialnet.unirioja.es/servlet/libro?codigo=926431

Franganillo, J. (2023). Generative artificial intelligence and its impact on the creation of media content. *Methaodos. Journal of Social Sciences, 11 (2), m231102a10.* https://doi.org/10.17502/mrcs.v11i2.710

Frazee, J.P. (2024, April 10). *Using Student Data to Bridge the AI Divide. EDUCAUSE Review.* https://er.educause.edu/articles/2024/4/using-student-data-to-bridge-the-IA-divide

García-Peñalvo, FJ (2023). *Open discussion on benefits, risks and challenges of Generative artificial intelligence.* CINIAC. https://doi.org/10.5281/zenodo.10029703

García-Peñalvo, F.J. (2024). Generative artificial intelligence and education: An analysis from multiple perspectives. *Education in the Knowledge Society, 25 (1), e31942.* https://doi.org/10.14201/eks.31942

García-Peñalvo, F.J. (2024). *Generative artificial intelligence in higher education: A 360° perspective.* University of Salamanca. https://doi.org/10.5281/zenodo.10499827

García-Peñalvo, F.J., Llorens-Largo, F., & Vidal, J. (2024). The new reality of education in the face of advances in generative artificial intelligence. *RIED. Ibero-American Journal of Distance Education, 27 (1), 1-25.* https://doi.org/10.5944/ried.27.1.37716

Giannini, S. (2023, July 6). Generative artificial intelligence in education: What are the advantages and disadvantages? Opportunities and challenges? *UNESCO.* https://unesdoc.unesco.org/ark:/48223/pf0000385877_eng

Grezan. (2024, October 4). Artificial intelligence in Latin American education: A call for reflection and collaboration. *Grezan Blog.* https://grezan.cl/artificial-intelligence-in-latin-american-education-a-call-for-reflection-and-collaboration/

Gutiérrez, KM (2023). Generative artificial intelligence: Disruption and challenges. *Journal*

Machine Translated by Google

Approaches, 4 (2), 57-82.
https://revistasdigitales.uniboyaca.edu.co/index.php/EFQ/article/view/1075/838

IGNITE Serious Play. (2024). *Applications of generative artificial intelligence in the education.* https://igniteseriousplay.com/applications-of-generative-artificial-intelligence-in-education/

National Institute of Educational Technologies and Teacher Training (INTEF). (2024). *Guide on the use of AI in the educational field.* Ministry of Education, Vocational Training and Sports. https://code.intef.es/wp-content/uploads/2024/07/Gu%C3%ADa-sobre-el-uso-de-la-IA-en-el-%C3%A1mbito-educational-INTEF_2024 .pdf

ISTE. (2023). *Bringing AI to School: Tips for School Leaders.* https://cms-live-media.iste.org/Bringing_IA_to_School-2023_07.pdf

Jara, I., & Ochoa, JM (2020). *Uses and effects of artificial intelligence in education.* Inter-American Development Bank (IDB). https://publications.iadb.org/es/fIAr-lac-ethical-and-responsible-adoption-of-artificial-intelligence-in-latin-america-and-the-caribbean

Jimenez Linares, L., Lopez-Gomez, J. A., Martin-Baos, J. A., Romero, F. P., & Serrano-Guerrero, J. (2023). ChatGPT: Reflections on the emergence of generative artificial intelligence in university teaching. *Proceedings of the Jenui, 8, 113-120.* https://rua.ua.es/dspace/bitstream/10045/137019/1/JENUI_2023_014.pdf

Kap, M. (2024, January 17). Critical analysis of generative artificial intelligence in education. *AMIDI.* https://www.amidi.org/inteligencia-artificial-educacion/

Klopfer, E., Reich, J., Abelson, H., & Breazeal, C. (2024). *Generative AI and K-12 Education: An MIT perspective.* MIT. https://mit-genIA.pubpub.org/pub/4k9msp17/release/1

Llorens-Largo, F., Vidal, J., & García-Peñalvo, FJ (2023). It's here, it's here, and no one can hide: *Generative artificial intelligence in education.* Aula Magna 2.0 Blog. https://cuedespyd.hypotheses.org/14389

León, S. (2024). Artificial intelligence (AI) in the development of soft skills. *Redacta Blog.* https://blog.redacta.me/artificial-intelligence-ia-in-the-development-of-soft-skills

Lopezosa, C. (2023). Generative artificial intelligence in scientific communication: Challenges and opportunities. *Journal of Research and Innovation in Health Sciences, 5(1), 1-5.* https://doi.org/10.46634/riics.211

Machine Translated by Google

Marr, B. (2020, September 28). 9 Soft Skills Every Employee Will Need in The Age of Artificial Intelligence (AI). *Forbes.* https://www.forbes.com/sites/bernardmarr/2020/09/28/9-soft-skills-every-employee-will-need-in-the-age-of-artificial-intelligence-AI/

Martínez, RL, Pérez González, MN, & Caro Morales, JL (2023). Intelligence artificial intelligence in education. *Digital Journal of Information Technology and Systems, 7 (1), 100-106.* https://doi.org/10.61530/redtis.vol7.n1.2023.136.100-106

Microsoft. (2024). 7 Ways AI Can Take Business Manufacturing and Administration to the Next Level. *News Center Latin America.* https://news.microsoft.com/en-us/7-ways-ai-can-take-business-management-and-production-to-the-next-level/

Moya, BA, & Eaton, SE (2023). Examining recommendations for the use of the Generative artificial intelligence with integrity from a teaching and learning lens. *RELIEVE, 29 (2), Art. M1.* https://doi.org/10.30827/relieve.v29i2.29295

Office of Superintendent of Public Instruction (OSPI). (2024). *Human-centered Artificial Intelligence in Schools.* OSPI. https://ospi.k12.wa.us/student-success/resources-subject-area/human-centered-artificial-intelligence-schools

Office of Superintendent of Public Instruction (OSPI). (2024a). *Human-centered AI guidance for K–12 public schools.* OSPI. https://ospi.k12.wa.us/sites/default/files/2024-08/comprehensive-IA-guidance.pdf

Office of Superintendent of Public Instruction (OSPI). (2024b). *Building AI Foundations: A Human-Centered Approach.* OSPI. https://ospi.k12.wa.us/sites/default/files/2024-07/IA-guidance_foundations.pdf

Office of Superintendent of Public Instruction (OSPI). (2024c). *Ethical Considerations for AI: A Framework for Responsible Use.* OSPI. https://ospi.k12.wa.us/sites/default/files/2024-06/IA-guidance_ethics.pdf

Office of Superintendent of Public Instruction (OSPI). (2024d). *Implementing AI: A Practical Guide for the Classroom.* OSPI. https://ospi.k12.wa.us/sites/default/files/2024-06/IA-guidance_classroom-considerations.pdf

Oliver, N. (2023). *Artificial intelligence, naturally: A manual for coexistence between humans and machines.* National Observatory of Telecommunications and the Information Society. www.ontsi.es/sites/ontsi/files/2020-06/Artificial IntelligenceNuriaOliver.pdf

Machine Translated by Google

Oregon Department of Education. (2023). *Developing Policy and Protocols for the Use of Generative AI in K-12 Classrooms.* Oregon Department of Education. https://www2.ed.gov/documents/IA-report/IA-report.pdf

Organization of Ibero-American States (OEI) & ProFuturo Foundation. (2023). *The future of artificial intelligence in education in Latin America.* ISBN 978-84-86025-22-9. https://oei.int/oficinas/secretaria-general/publicaciones/el-futuro-de-la-inteligencia-artificial-en-educacion-en-america-latina

Ramallo, H. (2024). *Generative AI Fundamentals for Teachers.* Udemy. https://www.udemy.com/course/fundamentos-de-ia-generativa-para-profesores

CAT Editorial. (2024). Soft skills and generative artificial intelligence. *Central Test.* https://aprendizajetransformacional.com/habilidades-blandas-inteligencia-artificial-generativa/

Sánchez Mendiola, M., & Carbajal Degante, E. (2023). Generative artificial intelligence and university education: Did the genie come out of the lamp? *Educational Profiles, 45 (Special), 70-81.* https://doi.org/10.22201/iisue.24486167e.2023.Special.61692

Sánchez Vera, M. del M. (2023). Artificial intelligence as a teaching resource: Uses and Opportunities for teachers. *Educar, 2023* (Article in press). https://doi.org/10.5565/rev/educar.1810

Santiago. (2024, May 24). Generative AI applications in education. *InnovIAtors.* https://www.innovIAtors.IA/generative-ai-applications-education

Shiohira, K. (2021). *Understanding the impact of artificial intelligence on skills development.* UNESCO. https://unesdoc.unesco.org/ark:/48223/pf0000376162

SkillCycle. (2024). *AI at work: The importance of soft skills in the workplace.* https://www.skillcycle.com/blog/IA-at-work-the-importance-of-soft-skills-in-the-workplace/

Skills You Need. (2024). *Soft skills in the era of AI.* https://www.skillsyouneed.com/rhubarb/soft-skills-IA.html

Staff. (2023, May 16). Artificial intelligence: How positive is it that AI comes into our lives? *TyN Magazine.* https://tynmagazine.com/artificial-intelligence-how-positive-is-it-that-AI-comes-into-our-lives/

Spencer, J. (2024). Promoting academic integrity in the age of generative AI. *John Spencer*

Machine Translated by Google

Education Blog. https://spencereducation.com/IA-academic-integrity/

Spencer, J. (2024a). *The 5 soft skills your organization needs in the age of AI.* https://corporate.britishcouncil.org/insights/5-soft-skills-your-organisation-needs-age-IA

Teach For America. (2023). The promises and perils of generative AI in education: TFA's evolving perspective. *Teach For America.* https://www.teachforamerica.org/stories/generative-IA-in-education

TrIAning Industry. (2023). *The AI advantage in soft skills trIAning.* https://trIAningindustry.com/articles/content-development/the-IA-advantage-in-soft-skills-trIAning/

UNESCO. (2022). *Recommendation on the ethics of artificial intelligence.* United Nations Educational, Scientific and Cultural Organization. https://unesdoc.unesco.org/ark:/48223/pf0000381137_eng

UNESCO. (2022). *Reimagining our futures together: A new social contract for the education.* United Nations Educational, Scientific and Cultural Organization. https://unesdoc.unesco.org/ark:/48223/pf0000379381_eng

UNESCO. (2024). *Global Education Monitoring Report 2023: Technology in Education. Education: A tool on whose terms?.* United Nations Educational, Scientific and Cultural Organization. https://unesdoc.unesco.org/ark:/48223/pf0000388894

UNESCO. (2024). *Guide to the use of generative AI in education and research.* United Nations Educational, Scientific and Cultural Organization. https://unesdoc.unesco.org/ark:/48223/pf0000389227

US Department of Education. (2023). *Artificial intelligence and the future of teaching and learning: Insights and recommendations.* Office of Educational Technology. https://www.ed.gov/sites/ed/files/documents/IA-report/IA-report.pdf

US Department of Education. (2024). *The Family Educational Rights and Privacy Act Guidance for parents.* https://studentprivacy.ed.gov/sites/default/files/resource_document/file/Ferpa-for-parents-spanish.pdf

Ducens University. (2024). Importance of generative artificial intelligence in the

Machine Translated by Google

Education: Implications and uses. *Ducens University.* https://universidadducens.edu.mx/blog/importance-of-artificial-generative-intelligence-in-education

Valenzuela Ramírez, S.G., Contreras Basurto, A., & Rivera Landeros, E.A. (2024). The introduction of artificial intelligence as an educational tool for teachers. *Ingenuity and Consciousness: Scientific Bulletin of the Sahagún City Higher School, 11 (22), 150-152.* https://doi.org/10.29057/escs.v11i22.12574

World Economic Forum. (2024). How AI is making soft skills more important in the workplace. https://www.weforum.org/agenda/2024/01/IA-value-soft-skills-workplace-jobs/

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Appendix. 5-Step Scaffolding Scale for Students in AI Integration

The Washington *Office of Superintendent of Public Instruction* (OSPI) published a set of documents on AI that includes: *Implementing AI: A Practical Guide for the Classroom (Washington OSPI, 2024d)24* which indicates that its content can be shared under a *Creative Commons Attribution License.* On page six of the cited document, the "5 Step Scaffolding Scale for Students" appears, for which the following translation is presented:

«The following 5-step implementation framework can be a practical tool to help students understand the various ways and levels in which AI can support their learning journey.

"This framework, designed to categorize AI integration from basic assistance to advanced interactive learning, serves as a roadmap for students to visualize how AI technologies can be progressively used in their education. Starting with basic AI assistance, such as personalized study aids, the ladder progresses through increasing levels of complexity, including collaborative problem solving with AI and culminating in creative projects that combine AI insights with human creativity. By introducing this ladder into the classroom, educators can guide students through each level, encouraging them to explore and experiment with AI tools that fit their current learning needs and aspirations. This approach not only helps students understand AI, but also empowers them to take control of their learning, recognizing AI as a valuable ally that can be adapted to support their educational goals through a variety of activities, from improving study habits to facilitating work on innovative projects."

Level 1 Without Assistance of AI	***Level 2 Assistance with AI Generation of ideas***	***Level 3 AI Supporting Drafting***	***Level 4 Collaborative AI Creation***	***Level 5 AI as Co-Creator***
				Human Inquiry HAH Human Empowerment
No AI tools are used in no way moment. *Students depend solely on their knowledge and skills.*	*AI tools can help generate ideas.* *The final content must be created by the student without direct AI input.* *AI assistance should be mentioned.*	*AI can help you write initial versions.* *The final version must be significantly revised by the student.* *There must be a clear distinction between the contribution of AI and that of the student.*	*AI-generated content may be included.* *The student must critically evaluate and edit the contributions of AI.* *The use of AI must be transparent and cited.*	*Extensive use of the AI in the creation of content.* *The student provides a justification for the use of AI and ensures original thinking.* *The work adheres to the academic integrity with appropriate citations.*

[24] Office of Superintendent of Public Instruction (OSPI). (2024d). *Implementing AI: A Practical Guide for the Classroom.* OSPI. https://ospi.k12.wa.us/sites/default/files/2024-06/ai-guidance_classroom-considerations.pdf

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